

ADI_SoundBar_SDK_SigmaStudio_Plus _Integration_Manual_EVAL-MELODY-8

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Preface

Thank you for purchasing the 21569SDK07 from Analog Devices, Inc. With a rich, powerful instruction set, floating point precision and high-speed execution, it makes SHARC processors a popular choice for designers of signal processing systems.

The ADI_SoundBar_SDK_SigmaStudio_Plus_Integration_Manual_EVAL-MELODY-8 is designed to be used in conjunction with the Cross Core Embedded Studio (CCES) development environment and GRIFFIN Ultra lite Reference Platform with an ADSP-21569 processor.

Access to the processor from a personal computer (PC) is achieved through an external JTAG emulator.

Analog Devices JTAG emulators offer faster communication between the host PC and target hardware. Analog Devices carries a wide range of in-circuit emulation products. To learn more about Analog Devices emulators and processor development tools, go to

<https://www.analog.com/en/resources/processors-and-dsp/processor-dev-tools.html>

Product Overview

The product features SoundBar software development environment supporting:

- HDMI input interface for high-definition audio Input
- Provision to develop and integrate Customer Post Processing in the SoundBar Audio signal chain.

Purpose of This Manual

The ADI_SoundBar_SDK_SigmaStudio_Plus_Integration_Manual_EVAL-MELODY-8.pdf provides information about SigmaStudio Plus integration, Design specification and Test specification.

Intended Audience

The SHARC SDK Manual is intended for use by software team and customers, both externally and internally to Analog Devices, to develop and port codes on ADSP-21569 Eval-Melody-8

Manual Contents

The manual consists of:

- Chapter 1, "Introduction"
Describes basic information covered in this document.
- Chapter 2, "Design Specification"
Provides details about the Hardware and Software Requirements, Software Architecture, Product Integration, Integration procedure and criteria.
- Chapter 3, "Test Specification"
Describes the Sigma Studio Plus testing procedure.

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This evaluation system supports Analog Devices 2156x processors.

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To access a complete technical library for each processor family, go to http://www.analog.com/processors/technical_library. The manuals selection opens a list of current

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1.Introduction

This document explains the integration of Sigma Studio Plus into 21569SDK07 real-time SDK. Sigma Studio Plus for SHARC includes an extensive set of algorithms to perform audio processing tasks such as filtering and mixing, as well as basic low-level DSP functions, optimized to run on the SHARC family of processors.

1.1 Purpose and Scope

The scope of the project is to integrate Sigma Studio Plus with Design and Deployment Mode into 21569SDK07 real-time SDK and validate its functionalities.

1.2 References

- AE_42_SS4G_Integration Guide.pdf
- AE_42_SS4G_QuickstartGuide.pdf
- AE_42_SS4G_ProductSpec.pdf
- AE_42_SS4G_HostcontrollerGuide.pdf
- Library Integration Code for Sigma Studio Plus 2.1.0

2. Design Specification

Codec	- ADSP21569SDK07
CCES Version	- 2.9.1.0
Target Processor	- ADSP21569.
	- Input channel- 16, 18, 24 Channel.
Channel Configuration	Output channel- Supports maximum 24 channels.
Input Data Format	- Channel based: Block Floating point
Output Data Format	- Floating-Interleaved
Block size	- 256

- In 21569SDK07 real-time SDK, Sigma Studio Plus is Integrated.
- For Sigma Studio Plus functionality both Design and Deployment Mode are integrated and tested with the Sigma Studio Plus schematics

2.1 Hardware Requirements

- EVAL Melody 8 and its accessories
- ICE 2000/1000
- Micro USB Data transfer cable
- EVAL_ADUSB2(USBi)
- DB cable & Speaker

2.2 Software Requirement

- CCES Version 2.9.1
- Sigma Studio Plus 2.1.0

2.3 Connections

This chapter explains connections of EVAL-MELODY-8 for Sigma Studio Plus.

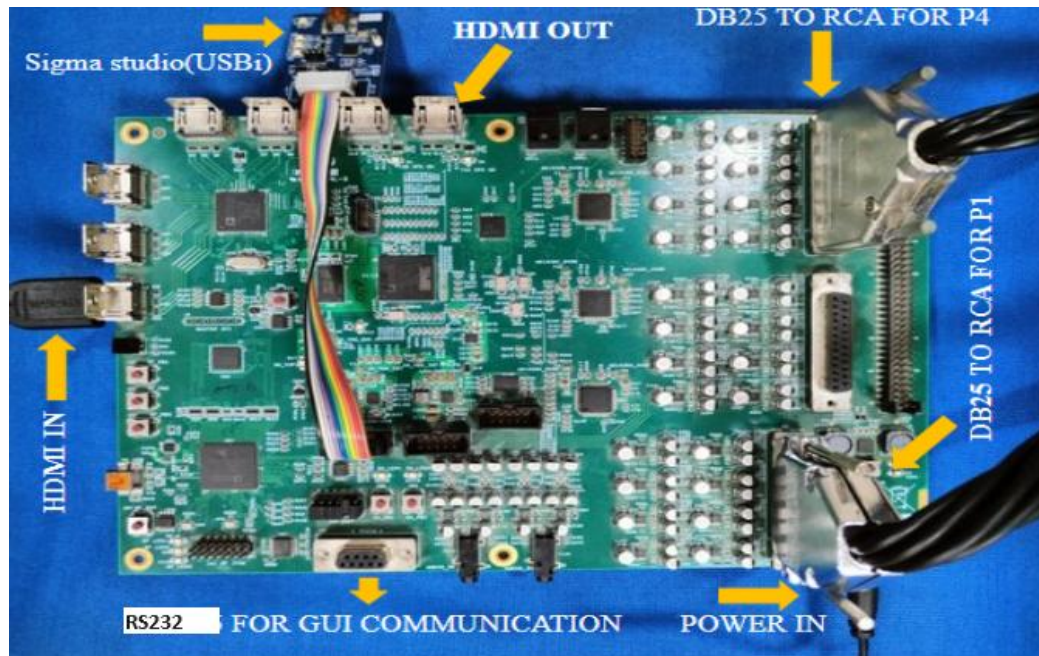


Figure 1 Overall connection of EVAL-MELODY-8

2.3.1 USBi for Sigma studio

1. Connect the Sigma studio USBi in port number 8(USBi 1).

Note:

- Remove the Resister R913 & R914 to use SPI1 in EVAL Melody 8.
- There are 2 versions of USBi one is new version (1.5v) and another one is old version (prev1.5v).
- For using the new version of USBi (1.5v), replace the resistors R765/6/7/8 & R1036/7/8/9 with zero-ohm resistors and remove R930 and R931 resistors. For using the old version of USBi, this hardware modification is not required.

2.4 Software Architecture

The software architecture of Sigma Studio Plus integrated is given below.

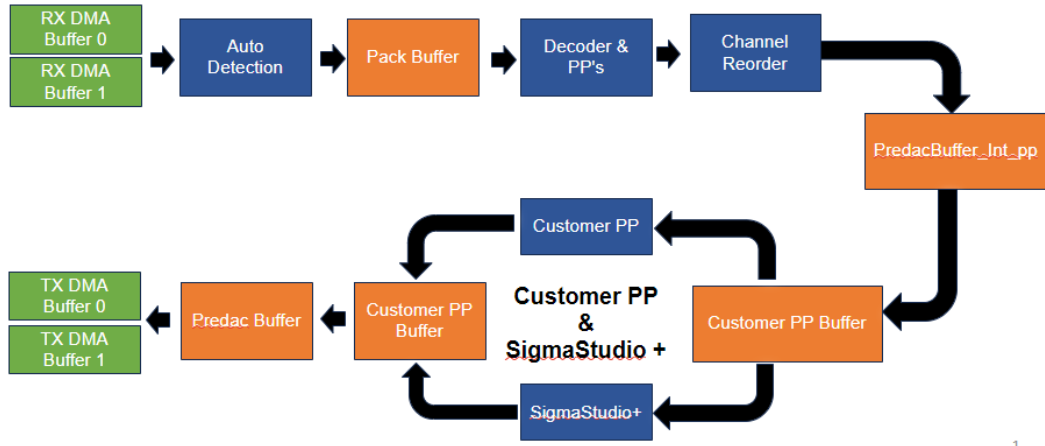


Figure 2 Buffer flow diagram of Sigma Studio Plus integrated in SDK

- As in the buffer flow diagram in Figure 1, the input PCM/Encoded audio is received in PackBuffer.
- The decoded output is copied to the PredacBuffer_Int_pp and the content is converted to floating-point data after the Post Processing routine.
- PredacBuffer_Int_pp_MAT holds the output from Post Processing routine. The Channel Reorder routine is used to route the decoded channel contents to the required position. The Channel Reorder routine copies the data in Customer PP buffer in the order L, R, C, Sw (Lfe), LS, RS, Assign Line 1, Assign Line 2, Assign Line3, in which, the channel position of L, R, C, Sw (Lfe), LS, RS is fixed in the Customer PP buffer.
- The floating-point data is copied to CustomerPP_Buffer and customer pp gain control API is call, and CustomerPP_Buffer which is passed to Sigma Studio Plus module if SSP_ENABLE is set 1. After Sigma Studio Plus processing or Customer Post Processing audio samples are converted to fixed point data and copied to PredacBuffer, and the PredacBuffer Fixed point data is sent for transmission.
- After completing the audio signal chain, the processed PCM samples are transmitted to DACs via SPORT Tx DMA buffers.

2.5 Product Integration

Sigma Studio Plus flowchart architecture,

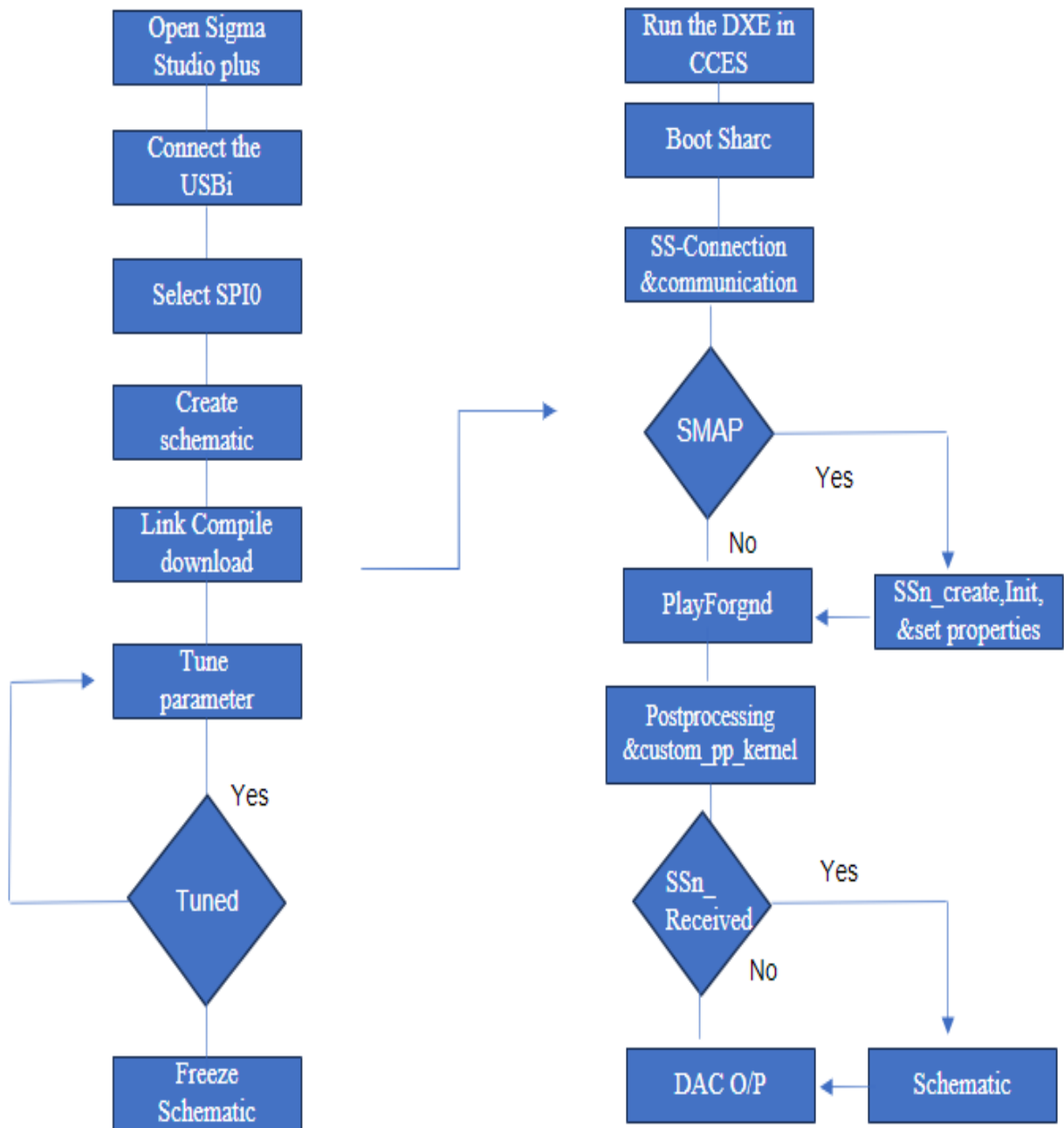


Figure 3 Sigma Studio Plus Design Mode flow chart

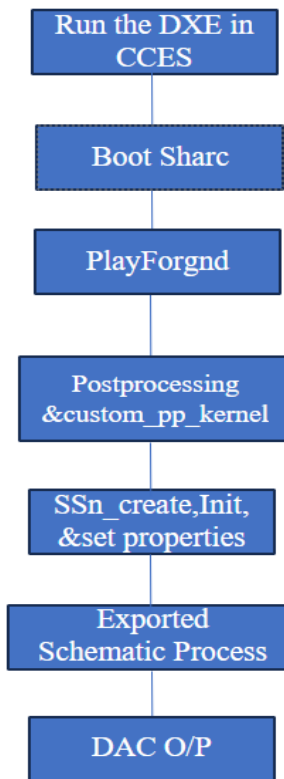


Figure 4 Sigma Studio Plus Deployment Mode flow chart

2.6 Integration procedure and criteria

The process of initializing and integrating Sigma Studio Plus involves several key steps.

- The integrated Sigma Studio Plus has two different Mode:
DesignMode
Deployment Mode
- The Sigma Studio Plus functionality can be enabled or disabled by setting it to 1 or 0 using the **SSP_ENABLE** macro.
- To enable design mode functionality the “**SSP_DESIGN_MODE**” macro is set to 1.
- To enable deployment mode functionality the “**SSP_DESIGN_MODE**” macro is set to 0.
- SPI1 pin is used for the communication between Sigma Studio Plus and SHARC (ADSP21569) processor. In CCES go to system.svc → pin multiplexing and select SPI1 option, where select CLK, MISO, MOSI and

SEL1 to SEL7 to establish the connection and communication for SigmaStudioPlus.

2.6.1 SigmaStudio Plus API

The API for SigmaStudio Plus is shown below,

SigmaStudio Plus can be added in the function "Customer_PP_kernel" in "Customer_PP.c" file.

```
void Customer_PP_kernel()
{
    CustPPInputList[0] = CustomerPP_Buffer;
    CustPPInputList[1] = Frmwk.SampleBlock_PP * CUSTOMER_PP_CHANNELS_CNT;
    CustPPInputList[2] = Frmwk.SampleBlock_PP;
    CustPPInputList[3] = CUSTOMER_PP_CHANNELS_CNT;
    CustPPInputList[4] = Frmwk.Customer_pp.Data_Format;
    CustPPInputList[5] = Frmwk.Customer_pp.ChannelMask;
    CustPPInputList[6] = Frmwk.StreamType;
    CustPPInputList[7] = Frmwk.OutputSampleRate;
    CustPPInputList[8] = Frmwk.Customer_pp.Volume;

    CustomerPP_gain_control(CustPPInputList);

    if(SSPLUS_Enable==1)
    {
        Sigmastudio_Plus_Process(CustomerPP_Buffer);
    }

    Customer_PP_Status_Update();
}
```

2.6.2 Step by Step Procedure for using Sigma Studio Plus controls.

The Sigma Studio Plus development environment provides a graphical user interface to help users build and deploy signal processing systems quickly and efficiently. It includes an extensive library of algorithms to perform audio processing such as filtering, mixing, and dynamics processing, as well as basic low-level DSP functions and control blocks. This environment implemented in two modes.

- Design mode
- Deployment mode

2.6.2.1 Sigma Studio Plus Memory segment in LDF file

- Segments placed in L1 Memory:
“SSPlus_SSncode”, “SSPlus_SIGMA_Design_Data”.
- Segments placed in L2 Memory: “SSPlus_SSnDataL2”.
- Segments placed in DDR Memory:
“mem_DMCO_SDRAM_SIGMA_CODE”, “SSPlus_SSnDataC”,
“SSplus_SSnDataB”, “SSPlus_SIGMA_Deploy_Data”,
“SSPlus_SSnData”, “SSPlus_SSnParameter”.

Design Mode:

In design mode we can access the schematics in Realtime.

- For design mode the Sigmastudio_Plus_init() function is invoked first, where the initial connection and communication for Sigma Studio Plus occur as a one-time process.
- Then the Sigmastudio_Plus_Config() function call is done in the Main.c above the PlayForGnd function where all the process for design mode is handled with the separate macros.
- Further the Postprocessing Routine is initiated where the Sigmastudio_Plus_Process() function is called.
- Once the SMAP is received, the adi_ss_create function creates the handler (hSSnHandle) and the SSn handle is populated in the Communication property's structure using the “adi_ss_comm_SetProperties” API.
- Subsequently, the Adi_SS_init is done, the aSSInOutBuff is cleared.
- Further in the PlayForGnd function, all the decoders' functions are called.

- Then the Postprocessing Routine is initiated where the `Sigmastudio_Plus_Process()` function is called to perform `adi_ss_schematic_process()`.

Deployment Mode:

In deployment mode the schematic is fixed in code and the module parameters in the schematics can be tuned using TTL (Host commands).

- For Deployment mode a schematic is fixed in Sigma Studio Plus and exported as .c and .h format for testing its functionalities.
- The `Sigmastudio_Plus_init()` function is invoked first, where the connection and communication are not done for deployment mode.
- Then the `Sigmastudio_Plus_Config()` function call is done in the `Main.c` above the `PlayForgrnd` function where all the process for deployment mode is handled with the separate macros.
- As an additional .C and .h file is included, all the SMAP details of the fixed schematic is obtained from `adi_ss_smap_DiffDXESchematic_0` array and accessed by `pSSnInfo` pointer.
- For this mode, the `adi_ss_create` function creates the handler (`hSSnHandle`), an mdma copy is done to receive all the code and parameter details from the `adi_ss_code_DiffDXESchematic_0`, `adi_ss_param_DiffDXESchematic_0` array available in the exported .c file.
- Subsequently, the `Adi_SS_init` is done, the `aSSInOutBuff` is cleared.
- Further in the `PlayForgrnd` function, all the decoders' functions are called.
- Then the Postprocessing Routine is initiated where the `Sigmastudio_Plus_Process()` function is called to perform `adi_ss_schematic_process()`.

2.6.3 Integration Sequence

Sigmastudio_Plus_init() Function: -

Here Sigma Studio Plus connection and communication will be done only for Design Mode.

Sigmastudio_Plus_Process() Function: -

Here Sigma Studio Plus schematic process will be done for both Design Mode and Deployment Mode.

2.6.4 Integration Environment

- CCES Version 2.9.1
- Sigma Studio Plus 2.1.0

2.7 Restrictions, Limitations and Constraints

- Make sure that Sigma Studio Plus macro “**SSP_ENABLE**” is set to 1 in the commn.h file.
- To perform with Design Mode, enable the “**SSP_DESIGN_MODE**” macro to 1 in the commn.h file.
- To perform with Deployment Mode, enable the “**SSP_DESIGN_MODE**” macro to 0 in the commn.h file.
- Ensure that all the hardware connections are made as per the figure in the user guide.
- Make sure that resistor R913 & R914 is removed from your melody 8 board (ADSP 21569) to use the SPI1.
- Replace the resistors R765/6/7/8 and R1036/7/8/9 with zero-ohm resistors and also remove R930, R931 resistors for using latest version USBi(v1.5).
- The “Firmware Version” property of the USBi in Sigma Studio Plus project should be set based on the version of the hardware.
- “Firmware Version” should be set as “v1.5” when the version of the USBi board is 1.5 or “Pre v1.5” should be used for 1.4 or earlier versions.
- For custom applications, please uncheck the target verification option in the Settings window of Sigma Studio Plus (Tools menu → Settings button → Processor Group → uncheck Target Verification checkbox) as shown in the below image.

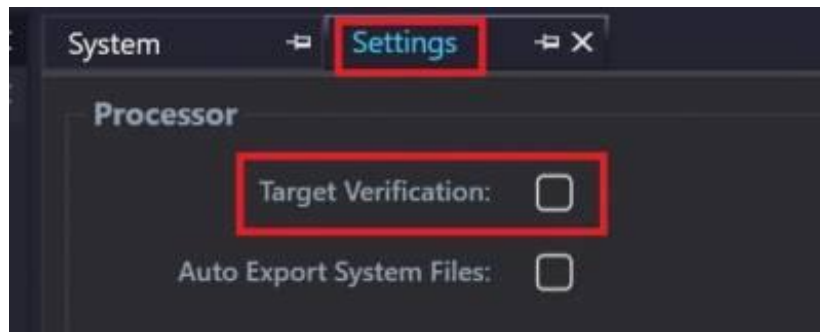
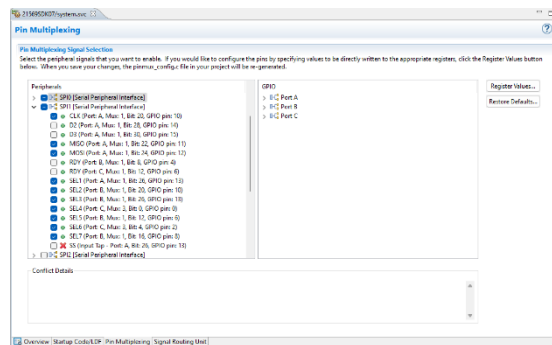


Figure 5 SSP Development Environment

- Make sure that when TX_DAC_Output_Ch_Cnt variable in the code is 3, 2 or 1 then the schematic must have input & output channel less than 24, 18, and 16 respectively.
- If input and output channel of the schematic is greater than TX_DAC_Output_Ch_Cnt variable, default audio output will be played. (For example, TX_DAC_Output_Ch_Cnt variable in the code is 1 (16channel) but the schematic input and output channel is set to 24 then the default audio will be the output.)
- SPI1 is used for the communication between Sigma Studio Plus and SHARC (ADSP21569) processor.
- In CCES go to system.svc → pin multiplexing and select SPI1 option, where select CLK, MISO, MOSI and SEL1 to SEL7 to establish the connection and communication for Sigma Studio Plus.



Pin Multiplexing

- In Design Mode the flag_configure and Flag_Toggle function is disabled since the flag 0 & SPI1 MOSI are on the same line (refer the figure below).

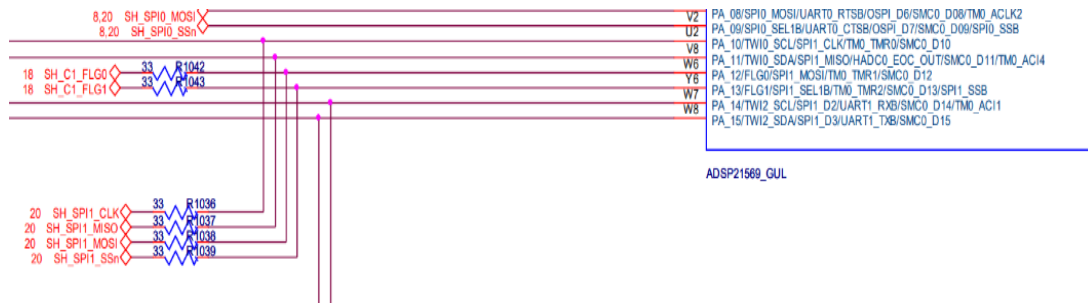


Figure 6 Flag configuration.

- For Deployment mode the flag_configure and Flag_Toggle function is enabled.
- The Sigma Studio Plus can support different block size but for integrated Design and Deployment mode the block size is fixed to 256. If the block size of the schematic is lesser than or greater than 256, default audio output will be played.
- For Deployment mode the Sampling Frequency is fixed to 48000.
- While working with deployment mode make sure that the current updated .dxs file must be considered for exporting .c and .h file.
- For Deployment mode the module parameters can be tuned using TTL (Host commands).
- TTL commands can be given from parameter 1 to parameter 30 where parameter 31 is reserved. Further if a few parameters are not required or not used, fill it with zeros till parameter 30. [For example: if parameters are only updated till parameter 28 then the last two must be filled with zeros (parameter 29 and parameter 30) as mentioned].
- While performing link compile download in default Code A segment is used for schematics in Sigma Studio Plus environment, in case if the schematic code memory exceeds the reserved memory size we can switch the code memory from Code A to Code B (L3) in Sigma Studio Plus GUI.
- Kindly open the Section Map settings and change the Code A (L1 memory) to CodeB (L3 memory) [EV21569SOM_0 → ADSP2156X_0 → SharcXICore_C1 → Schematic_C1_IO → Section Map → seg_swco → CodeB] to overcome the memory insufficient error.

Code Memory Sections

Code Section	Buffer
seg_swco	Code B

State Memory Sections

State Section	Buffer
seg_dmda1	State A
seg_dmda2	State A
seg_dmda3	State A

Figure 7 Code Memory Section

3. Test Specification

3.1 Sigma Studio Plus testing procedure

For Design Mode:

The Sigma Studio Plus (2.1.0) development environment consists of different windows and panels used to design basic building blocks to construct the project.

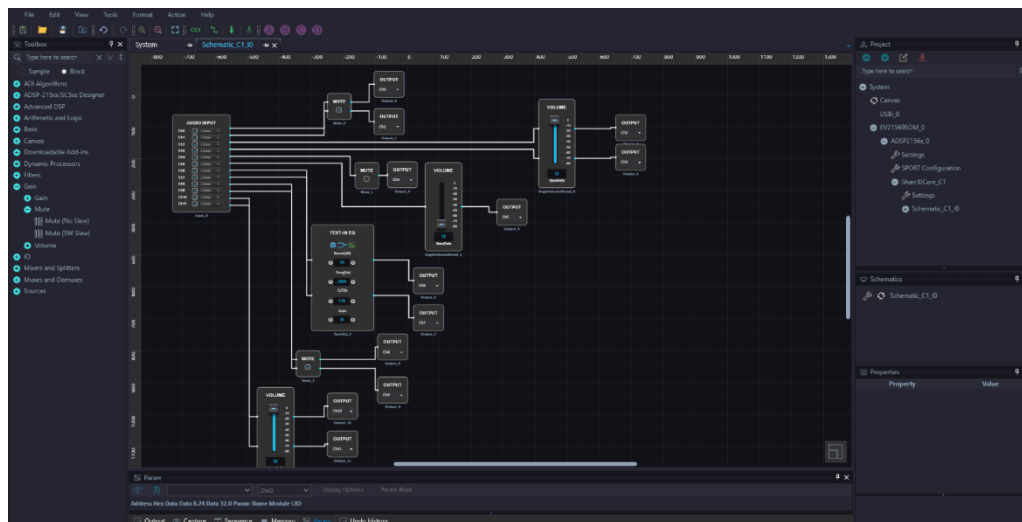


Figure 8 SSP Development Environment

- Drag the required modules to the Tabbed Document Panel for designing the required Schematic.

Execute an Example Project

- Target application must be loaded on to the evaluation platform before executing the example applications on ADSP-21569 platforms. Using Cross Core Embedded Studio load the application DXEs for the core through debug session using debug agent (ICE-1000/ICE-2000).
- Open Sigma Studio Plus application.
- Click the file menu to create a new project.

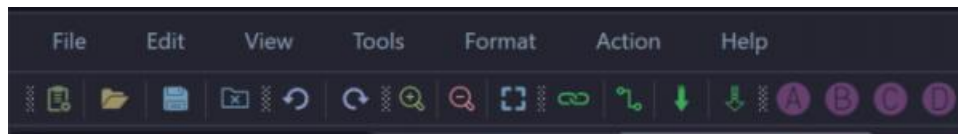


Figure 9 SSP Menu bar

- Whenever a new project is created, the first window appears is Tabbed Document Panel.
- Drag and drop the communication interface and connect to PC block also connect 21569 SHARC platform to the USBi as shown in the figure below.
- The figures below also show the tree toolbox window.

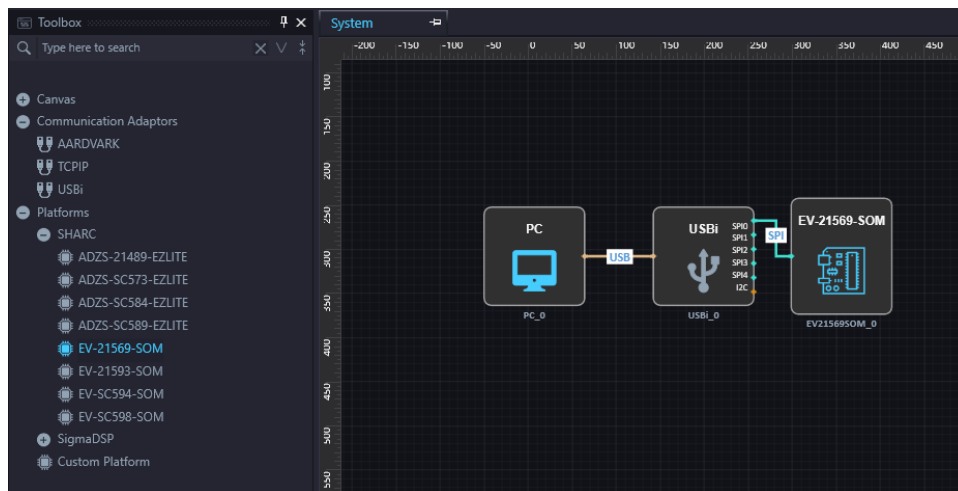


Figure 10 Toolbox window

- The “Firmware Version” property of the USBi in Sigma Studio Plus project should be set based on the version of the hardware.
- “Firmware Version” should be set as “v1.5” when the version of the USBi board is 1.5 or “Pre v1.5” should be used for 1.4 or earlier versions.

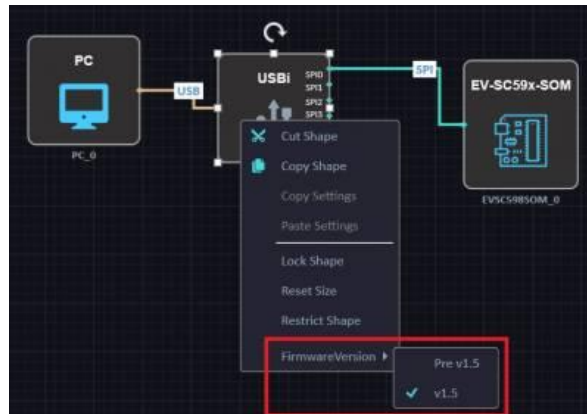


Figure 11 Firmware Version

- Double click on the EV21569SOM processor it will go to the Schematic_C1_I0 design window.
- Drag and drop blocks from the Toolbox panel into the schematic tab (Tabbed Document Panel) to create your design.

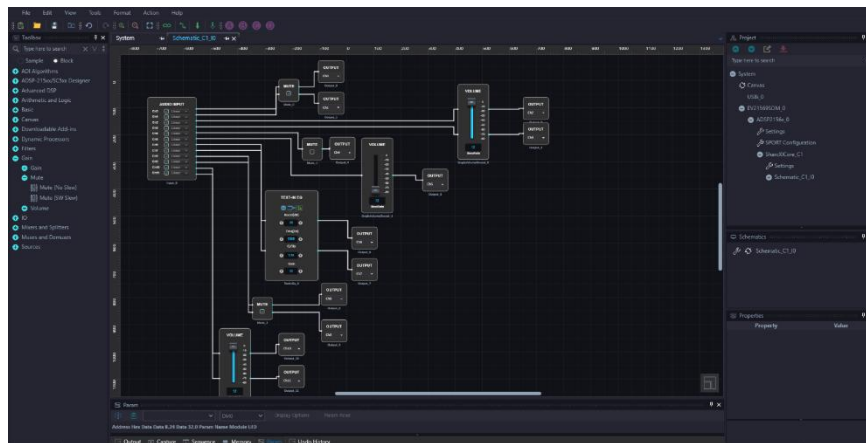


Figure 12 Schematic Design window

- Select the corresponding application executables (DXE's) for each of the cores in SHARC Core settings window (refer the below image).

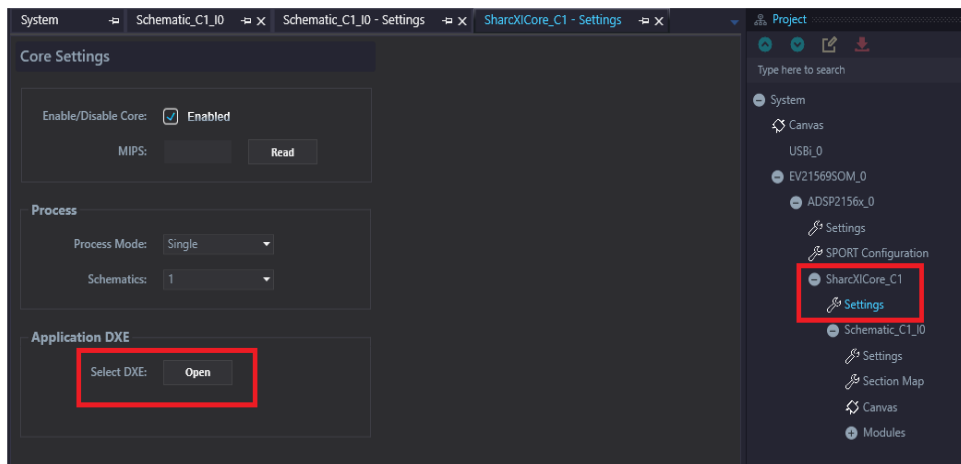


Figure 13 Core settings window

- Kindly Verify that the check box of core settings is enabled. Click the Read button the MIPS value of Sigma Studio Plus + schematics will be displayed (refer the below image).

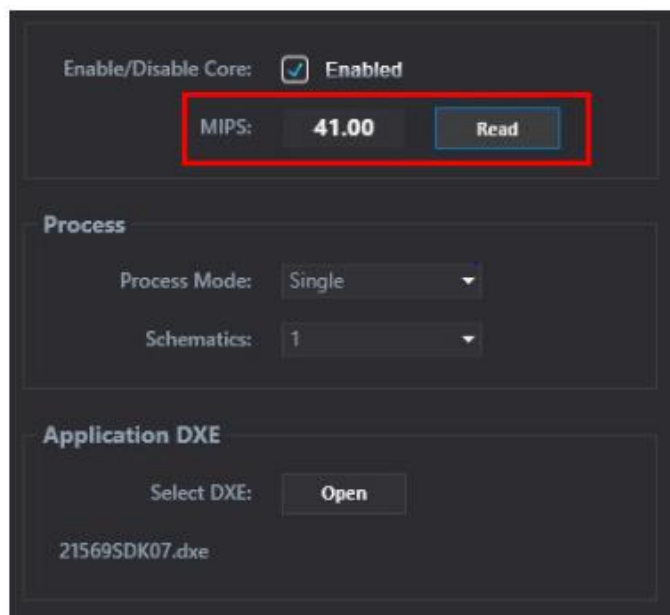


Figure 14 MIPS Calculation

- Open ADSP2156x_0 – Settings and for build mode select debug and select CCES version as 2.9.1 as per the reference image below.

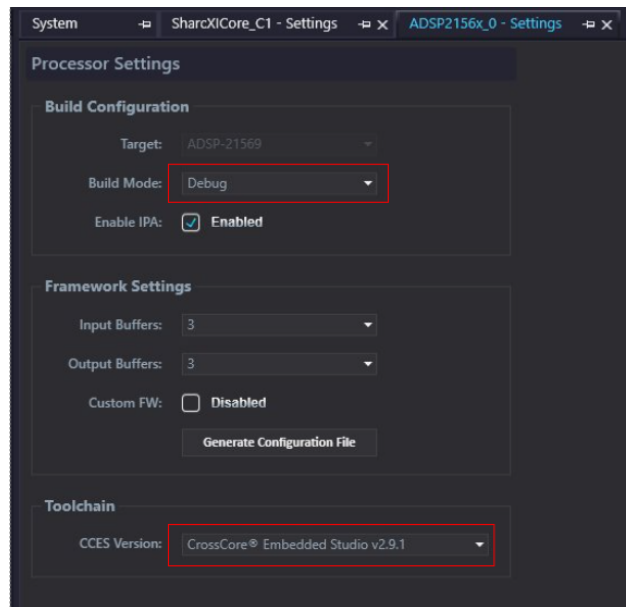


Figure 15 ADSP2156x_0 - settings window

- As per the reference image below, in the schematic settings page the number of inputs, output channels and the Block size for processing can be assigned.



Figure 16 Channel & Block size configuration

- Ensure that the Block size of the modules used in the tabbed document panel is also 256 same as the Block Size in the Schematic settings as per the reference image below.

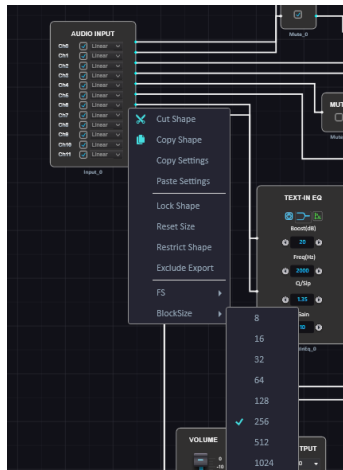


Figure 17 Block size

- For custom applications, please uncheck the target verification option in the Settings window of Sigma Studio Plus (Tools menu → Settings button → Processor Group → uncheck Target Verification checkbox) as shown in the below image.

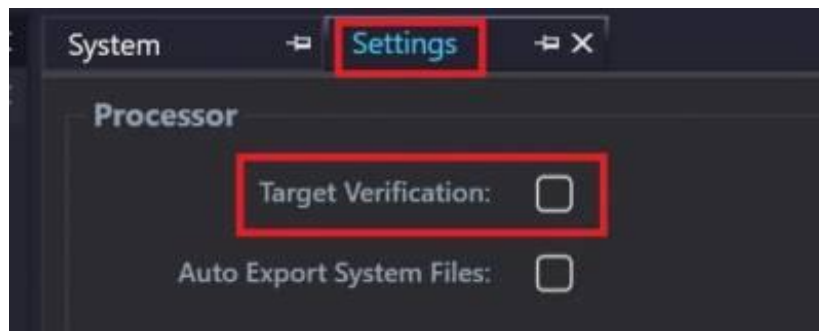


Figure 18 Target Verification

- Once the schematic design is done, you can link and compile the project using the down arrows as highlighted in the below image.

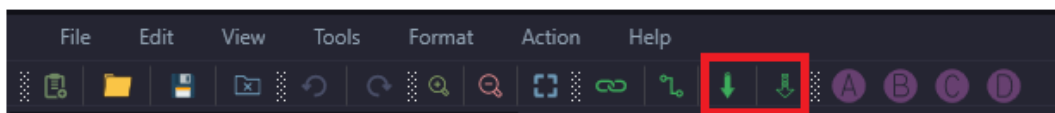


Figure 19 Link compile download

- After clicking the “link compile download”, once the schematic download is successful it will be displayed in the output window.

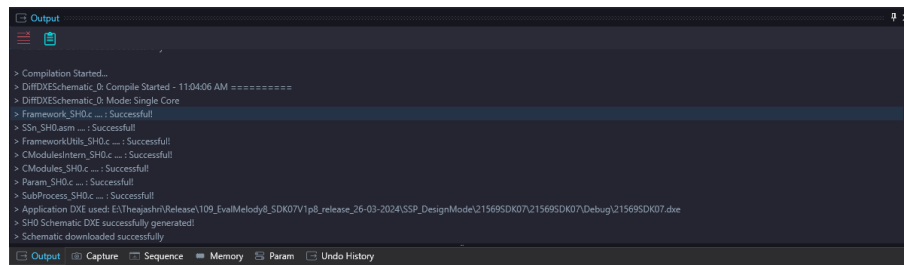


Figure 20 Output window

- If any errors are encountered during compilation, the error information will be displayed on the Output Window.
- Reference link for designing in Sigma Studio Plus environment:

<https://wiki.analog.com/resources/tools-software/sigmastudiov2>

NOTE:

- Make sure that the “**SSP_ENABLE**” and “**SSP_DESIGN_MODE**” is set to 1 for Sigma Studio Plus design mode process if not default User PP code will be loaded.
- In the core settings the built application .dx file path must be included to proceed with Link Compile Download operation.

For Deployment Mode:

1. In this mode the Code B in Sigma Studio Plus GUI is not supported.
2. As per above mentioned steps design a schematic, then select the updated .dx file and give link compile download in Sigma Studio Plus.
3. Once the ‘LinkCompile-Download’ action is complete, the system files can be exported by selecting Action → Export System Files from the toolbar as highlighted in the below image.

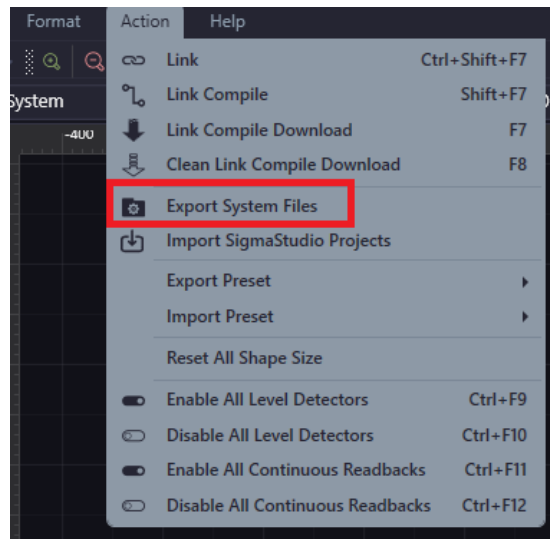


Figure 21 Export system

- 4.Note: The Action → Export System Files menu item is inactive if the Schematic is not compiled after any modifications.
- 5.The name of system files can be decided by the user.
- 6.The files contain code, parameter and version information to be sent to the SHARC Target. The files should be freshly generated every time the Schematic is modified.

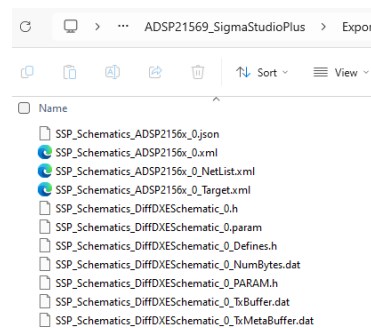


Figure 22 System files

7.These files, which can be opened in any text editor for analysis, that are essential for finding parameters and addresses to integrate into code. These files will help you tune the module via TTL using that address offset.

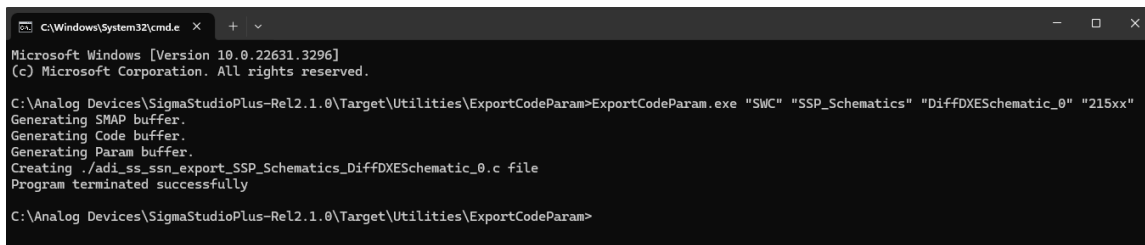
8.Refer AE_42_SS4G_HostControllerGuide for information about all files.

9.Copy all the Sigma Studio Plus exported files into “C:\Analog Devices\SigmaStudioPlus-Rel2.1.0\Target\Utilities\ExportCodeParam” folder.

10. A utility (ExportCodeParam.exe) is provided with the installer to do this formatting. Run the “exportCodeParam” utility as per commands shown in below.

11. Run the utility as below: ExportCodeParam.exe<Visa> <Export File name><lc_name><Processor type> For e.g. **ExportCodeParam.exe "SWC" "SSP_Schematics" "DiffDXESchematic_0" "215xx"**

Note: Export file name decided by the user



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22631.3296]
(c) Microsoft Corporation. All rights reserved.

C:\Analog Devices\SigmaStudioPlus-Rel2.1.0\Target\Utilities\ExportCodeParam>ExportCodeParam.exe "SWC" "SSP_Schematics" "DiffDXESchematic_0" "215xx"
Generating SMAP buffer.
Generating Code buffer.
Generating Param buffer.
Creating ./adi_ss_ssn_export_SSP_Schematics_DiffDXESchematic_0.c file
Program terminated successfully

C:\Analog Devices\SigmaStudioPlus-Rel2.1.0\Target\Utilities\ExportCodeParam>
```

Figure 23 Utility commands

12.This generates C formatted source and header file in the same folder.

13. Copy the generated ‘.c’ and ‘.h’ files to our project workspace .c file should be placed in “src/system/” folder and .h files should be placed in “src/system/include” folder as per below image.

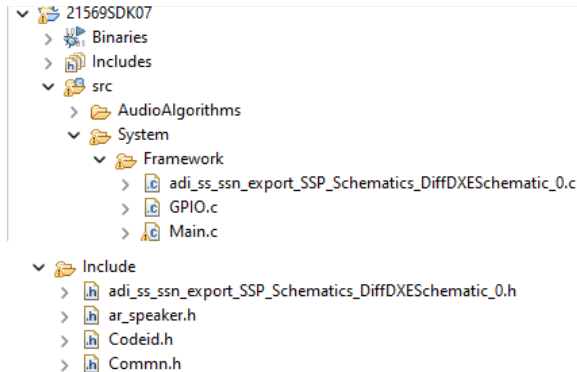


Figure 24 Add the .c and .h files

14.Launch Cross Core Embedded Studio on the Sigma Studio Plus Host PC. Specify the workspace when prompted.

15.Whenever exported, ensure that **exported .h header file name** is same in the **SS_Plus.c** file as per below image.

```

28 #include <sru21569.h>
29
30 #include "SS_Plus.h"
31 #include "adi_ss_common.h"
32 #include "Extern_C.h"
33 #include "adi_ss_ssn_export_SSP_Schematics_DiffDXESchematic_0.h"
34 #include "adi_ss_fw_dataHandle_Sh.h"

```

Figure 25 Header file name change

16.Rebuild the application for SHARC core and flash the ldr.

17.The output audio will be heard through speakers/headphones connected to the Board as per the connections in the schematic.

18.In deployment Mode the fixed schematic consists of 12 channel pass-through with mute module.

19.To Mute a specific channel TTL commands can be passed through RS232 port in ADSP21569

NOTE:

- Make sure that the “**SSP_ENABLE**” is set to 1 and “**SSP_DESIGN_MODE**” is set to 0 for Sigma Studio Plus deployment mode process if not default User PP code will be loaded.
- In the core settings the built application .dxs path must be included to proceed with Link Compile Download operation and export the schematic files.

TTL command specifications

1. The Command Byte for Sigma Studio Plus Control is specified as 0x48.
2. To tune a particular module of fixed schematic Module_ID is given as the first parameter command and the number of module values to be tuned (NumParam) is given as the second parameter command in the TTL.
3. The module value and the Module_ID can be referred from the Sigma Studio Plus parameter window. (For example: Module Id, Module value for a Mute (Module) in Parameter window is highlighted in the below

image).

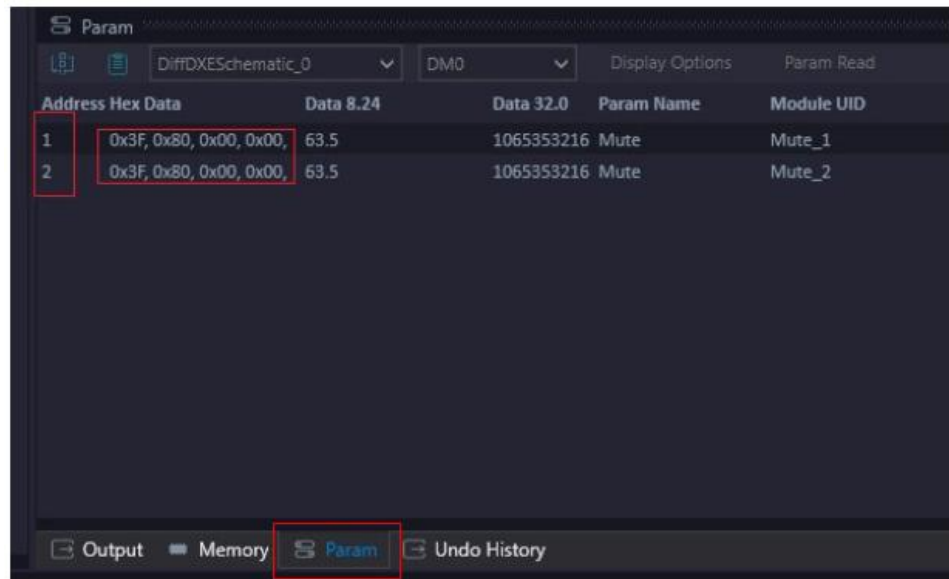


Figure 26 Module Id and Module Value

4. To tune the module values of a specific module, two parameter commands in TTL are considered (For example: Parameter 3 and Parameter 4).
5. The tunable values will be in hex format.
6. The LSB & the MSB value is given as next TTL command (ex: module value is 0x3F800000: Parameter 3: 0000 (LSB) and Parameter 4: 3F80 (MSB)).
7. To tune further modules in the same TTL again provide the Module_ID of that specific module as one TTL parameter command and the number of module value to be tuned (NumParam) as the next parameter followed by the module's LSB and MSB values sequentially.
8. TTL parameter commands can be given from 1 to 30 where, parameter command 31 is reserved. If any TTL parameter commands are not required or not filled, fill it with zeros till TTL command 30. [For example: if parameter command are only updated till 28 in TTL then the last two must be filled with zeros (29th and 30th TTL command) as mentioned].
9. If single module with more than 14 module values need to be tuned, the first 14 module values are updated using first TTL command then further

module values can be updated using the next TTL file (no need to provide the same Module_ID, NumParam again in the second TTL file).

Note:

- One module with “N” number of module values can be tuned.
- More than one module with “N” number of module values can be tuned.
- To tune a module value, LSB and MSB are given in two separate Parameter commands via TTL.

Example

Case1: More than one module with “N” number of module values can be tuned.

- An example schematic for 12 channel with gain, volume control and mute modules are designed.

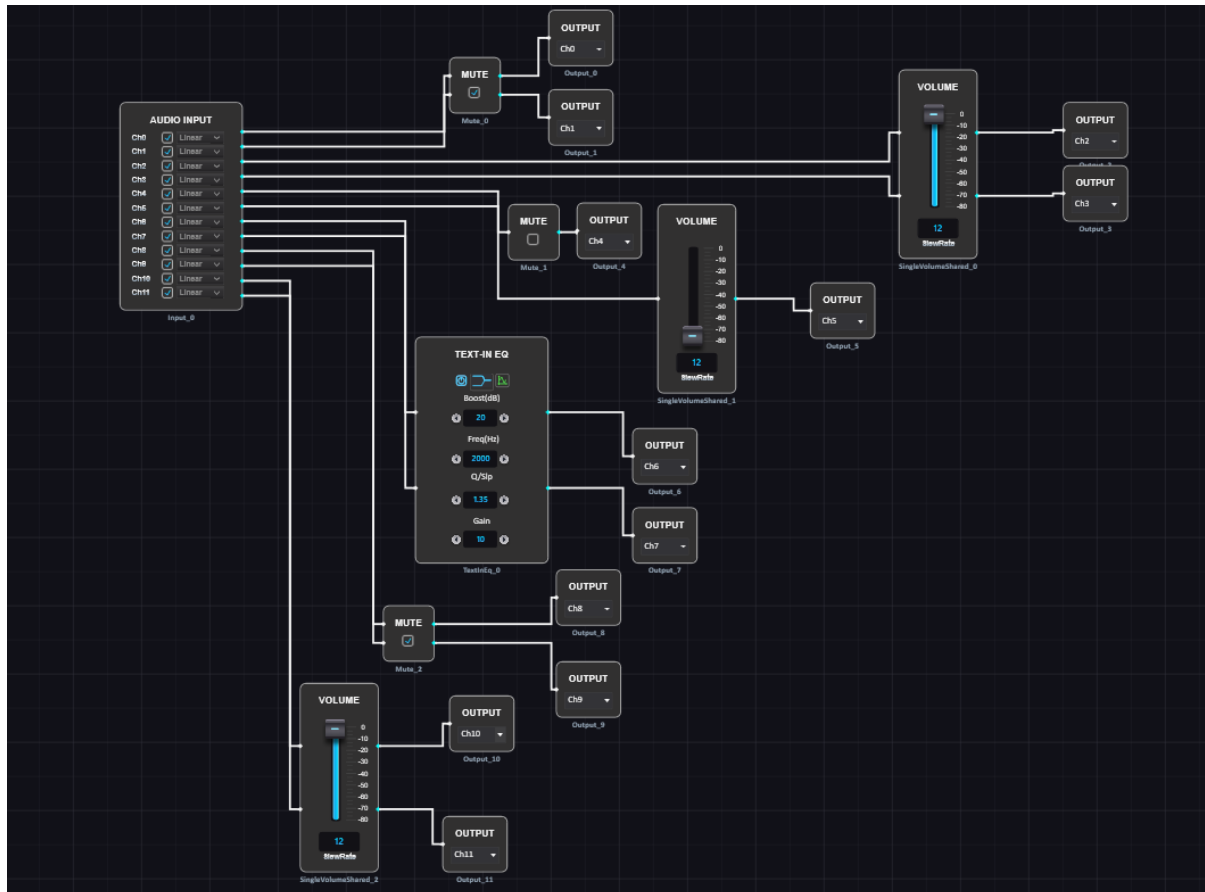


Figure 27 Schematic

- The module value and the Module_ID can be referred from the Sigma Studio Plus parameter window.

	Address Hex Data	Data 8.24	Data 32.0	Param Name	Module UID
1	0x39, 0x80, 0x00, 0x00,	57.5	964689920	SlewRate	SingleVolumeShared_2
2	0x3F, 0x64, 0x29, 0x05,	63.3912509083748	1063528704	Gain	SingleVolumeShared_2
3	0x00, 0x00, 0x00, 0x00,	0	0	Mute	Mute_2
4	0x40, 0x17, 0x42, 0xF2,	64.0908652544022	1075266304	Stage0_B2	TextInEq_0
5	0xC0, 0xB0, 0x9F, 0x4A,	-63.3100694417953	3232800512	Stage0_B1	TextInEq_0
6	0x40, 0x72, 0xF3, 0x6D,	64.4490268826485	1081275264	Stage0_B0	TextInEq_0
7	0xBF, 0x5B, 0x53, 0x02,	-64.6432646512985	3210433280	Stage0_A2	TextInEq_0
8	0x3F, 0xEB, 0x12, 0x32,	63.9182463884354	1072370240	Stage0_A1	TextInEq_0
9	0x39, 0x80, 0x00, 0x00,	57.5	964689920	SlewRate	SingleVolumeShared_1
10	0x39, 0x3A, 0x77, 0x53,	57.228383243084	960132928	Gain	SingleVolumeShared_1
11	0x3F, 0x80, 0x00, 0x00,	63.5	1065353216	Mute	Mute_1
12	0x39, 0x80, 0x00, 0x00,	57.5	964689920	SlewRate	SingleVolumeShared_0
13	0x3F, 0x80, 0x00, 0x00,	63.5	1065353216	Gain	SingleVolumeShared_0
14	0x00, 0x00, 0x00, 0x00,	0	0	Mute	Mute_0

Figure 28 Parameter window

TTL preparation:

1. The general decoding command is given first and if there is any Decoder + Post Processing then the respective Decoder commands need to be given.
2. Sigma studio plus command options needs to be given, data for each modules needs to be given as follows
3. Module ID (address) is given first followed by the no. of. parameters, LSB and MSB. In case of multiple parameters, the least module id can be maintained as default Module ID. For every module the TTL instructions are given as below.

Note: 1. Module ID should be in Hexadecimal format.

2. Only 30 parameters can be sent at once through TTL for more Channels separate TTL can be given with the same procedure.
3. If few parameters are not used or not filled, fill it with zeros till parameter 30.

Module Name = Mute_0
Module ID (Address) = 14
No.of.parameter = 1 (mute)
Address Hex Data = 0x00, 0x00, 0x00, 0x00

Module Name = SingleVolumeShared_0
Module ID (Address) = 12 & 13
No. of. parameter = 2 (slewRate & gain)
Address Hex Data = 0x39, 0x80, 0x00, 0x00 (SlewRate)
0x3F, 0x80, 0x00, 0x00 (Gain)

Module Name = Mute_1
Module ID (Address) = 11
No.of.parameter = 1 (mute)
Address Hex Data = 0x3F, 0x80, 0x00, 0x00

Module Name = SingleVolumeShared_1
Module ID (Address) = 9 & 10
No. of. parameter = 2 (slewRate & gain)
Address Hex Data = 0x39, 0x80, 0x00, 0x00 (SlewRate)
0x39, 0x3A, 0x77, 0x53 (Gain)

Module Name = TextInEq_0
Module ID (Address) = 4, 5, 6, 7 & 8
No. of. parameter = 5 (Stage buffers)
Address Hex Data = 0x40, 0x17, 0x42, 0xF2 (Stage0_B2)
0xC0, 0xB0, 0x9F, 0x4A (Stage0_B1)
0x40, 0x72, 0xF3, 0x6D (Stage0_B0)
0xBF, 0x5B, 0x53, 0x02 (Stage0_A2)
0x3F, 0xEB, 0x12, 0x32 (Stage0_A1)

Module Name = Mute_2
Module ID (Address) = 3
No.of.parameter = 1 (mute)
Address Hex Data = 0x00, 0x00, 0x00, 0x00

Module Name = SingleVolumeShared_2
Module ID = 1 & 2
No. of. parameter =2 (slewRate & gain)
Address Hex Data = 0x39, 0x80, 0x00, 0x00 (SlewRate)
0x3F, 0x64, 0x29, 0x05 (Gain)

```

sendln '0x2001000E' //Module ID -Mute_0
mpause 2000
sendln '0x20020001' //No of parameters
mpause 2000
sendln '0x20030000' //LSB
mpause 2000
sendln '0x20040000' //MSB
mpause 2000

sendln '0x2005000C' //Module ID -SingleVolumeShared_0
mpause 2000
sendln '0x20060002' //No of parameters
mpause 2000
sendln '0x20070000' //LSB
mpause 2000
sendln '0x20083980' //MSB
mpause 2000
sendln '0x20090000' //LSB
mpause 2000
sendln '0x200A3F80' //MSB
mpause 2000

sendln '0x200B000B'
mpause 2000
sendln '0x200C0001'
mpause 2000
sendln '0x200D0000'
mpause 2000
sendln '0x200E3F80'
mpause 2000

sendln '0x200F0009'
mpause 2000
sendln '0x20100002'
mpause 2000
sendln '0x20110000'
mpause 2000
sendln '0x20123980'
mpause 2000
sendln '0x20137753'
mpause 2000
sendln '0x2014393A'
mpause 2000

sendln '0x20150003'
mpause 2000
sendln '0x20160001'
mpause 2000
sendln '0x20170000'
mpause 2000
sendln '0x20180000'
mpause 2000

sendln '0x20190001'
mpause 2000
sendln '0x201A0000'
mpause 2000
sendln '0x201B3980'
mpause 2000
sendln '0x201C2905'
mpause 2000
sendln '0x201D3F64'
mpause 2000
sendln '0x201E0000'
mpause 2000
sendln '0x20000048'
mpause 4000

sendln '0x20010004'
mpause 2000
sendln '0x20020005'
mpause 2000
sendln '0x200342F2'
mpause 2000
sendln '0x20044017'
mpause 2000
sendln '0x20059F4A'
mpause 2000
sendln '0x2006C0B0'
mpause 2000
sendln '0x2007F36D'
mpause 2000
sendln '0x20084072'
mpause 2000
sendln '0x20095302'
mpause 2000
sendln '0x200ABF5B'
mpause 2000
sendln '0x200B1232'
mpause 2000
sendln '0x200C3FEB'
mpause 2000
sendln '0x200D0000'
mpause 2000
sendln '0x200E0000'
mpause 2000
sendln '0x200F0000'
mpause 2000
sendln '0x20100000'
mpause 2000
sendln '0x20110000'
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mpause 2000
sendln '0x20180000'
mpause 2000
sendln '0x20190000'
mpause 2000
sendln '0x201A0000'
mpause 2000
sendln '0x201B0000'
mpause 2000
sendln '0x201C0000'
mpause 2000
sendln '0x201D0000'
mpause 2000
sendln '0x201E0000'
mpause 2000
sendln '0x20000048'
mpause 4000

```

Figure 29: Example TTL Command

MIPS Details

Speaker Configuration	MIPS for Sigma Studio Plus	Free MIPS Available
7.1.4	40	85
5.1.2	30	200

Table 1 MIPS Details

Memory Details

L1 MEMORY	Purpose of the Segment	Size in HEX	Size in Mbits
SSPlus_SSncode	Reserved for Schematic Code	85FF	0.261711
SSPlus_SIGMA_Design_Data	Design Mode Framework Data	17B1	0.046272
L2 MEMORY	Purpose of the Segment	Size in HEX	Size in Mbits
SSPlus_SSnDataL2	Reserved for Schematic L2 Data	A40F	0.320427
DDR MEMORY (L3)	Purpose of the Segment	Size in HEX	Size in Mbits

mem_DMC0_SDRAM_SIGMA_CODE	Design Mode Framework Code and Code B	BF371	5.975471
SSPlus_SSnDataC	Reserved for Schematic Data C	27958	1.237
SSplus_SSnDataB	Reserved for Schematic Data B	1869F	0.762932
SSPlus_SIGMA_Deploy_Data	Deployment Mode Framework data	1EB16	0.959152

SSPlus_SSnData	Reserved for Schematic Data A	2795F	1.237053
SSPlus_SSnParameter	Reserved for Schematic Parameter	1869F	0.762932

Table 2 Memory Details